SAMPLE QUANTITIES REPORT

Finishing quantities

Example house: G+1, 1,000 sq ft per floor

PREPARED FOR	LOCATION	REPORT ID	DATE
Example house (sample)	Lucknow, Uttar Pradesh	NS-IN-SAMPLE	26 October 2026

This is a sample report
It is a real report from the same calculations a paid report uses, made for an example house (G+1, 1,000 sq ft on each floor, Lucknow). It is not your project and nothing in it applies to your plot. Your own report uses your plot, floors and city, so its figures will differ.

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Your finishing at a glance

ROOMS COUNTED
9
3 bedrooms, 3 bathrooms, 1 kitchen, 2 living rooms

BUILDING
One house
2 floors, 3.00 m floor to floor

PAINT INSIDE
Standard
Cement putty (about 1 mm), one coat of primer, two coats of emulsion

PAINT OUTSIDE
Acrylic emulsion
Smooth acrylic exterior emulsion, 2 coats over exterior primer

FLOOR TILES
600 × 600 mm
Laid straight, fixed with a mortar bed

BATHROOM WALLS
Tiled to the door height
300 × 600 mm wall tiles

FALSE CEILING
None
Gypsum board on a steel frame

KITCHEN
L shape
Counter only

WARDROBES
None
Built in, plywood and laminate

What to order

The main totals. The line under each figure is the range the real number is likely to fall in.

PAINTED AREA	TILES	DOORS AND WINDOWS	WINDOW GLASS
942 m²	254 m²	22 nos	16 m²
801 to 1,130 m²	228 to 292 m²	19 to 28 nos	13 to 21 m²

How to use this report
This is a code-based planning estimate worked out from the answers you gave, not a design for your drawings. Use it to plan, to check a contractor's quantities for gross errors and to order the first lots. Where a figure rests on an assumption, the report says so. Final quantities come from the drawings and the professional who signs them.

Surfaces, measured

Every area comes from the room plan, measured following the rules of IS 1200 as the tool reads them (editions named under Standards and sources; the 2024 revisions have not been checked). Walls are the clear height of the room, floor to the underside of the slab.

SURFACE	MEASURED (M²)	TAKEN OUT (M²)	NET (M²)	BASIS
Walls in rooms and passages	404.0	20.5	383.5	Paint; doors out once, windows half; the wall above the kitchen counter is tiled
Bathroom walls above the tiles	21.6	0.0	21.6	Paint above the 2.1 m tiling
Ceilings, inside	162.4	0.0	162.4	Slab or board, painted either way
Outside of the building	384.8	10.4	374.4	Walls, parapet, balcony edges and undersides; openings half from each face
Doors, both faces	33.2	0.0	33.2	Shutters of timber; frames are counted with them
Bathroom wall tiles	60.5	5.0	55.4	Door opening taken out
Kitchen wall above the counter, tiles	2.5	0.0	2.5	4.2 m run, 0.6 m high
Floors to be tiled	166.9	0.0	166.9	Measured before the skirting goes on (IS 1200 Part 11)

Clear height is 2.85 m (3.00 m floor to floor less a 0.15 m slab). Rooms are the same ones MasonryPro uses, so the two reports measure the same walls.

Putty, primer and paint

Standard: putty, primer and emulsion. Outside: smooth acrylic exterior emulsion, 2 coats over exterior primer. Quantities include 5% for wastage.

ITEM	AREA (M²)	SPREAD, ALL COATS, A M²	QUANTITY	UNIT
Wall putty, cement based	567.6	0.60 kg	358	kg
Primer, inside	567.6	0.075 L	45	L
Interior emulsion, 2 coats	567.6	0.090 L	54	L
Primer, outside	374.4	0.095 L	37	L
Exterior acrylic emulsion	374.4	0.167 L	66	L
Primer, doors	33.2	0.075 L	3	L
Enamel or multi-surface paint, doors	33.2	0.090 L	3	L

Spread is for the coats named, on a plastered surface. Buy paint in the pack sizes you like and round up; keep a litre or two for touching up. CPWD DAR 2016 item 13.46.1 (0.167 L a m²); primer from the maker sheet, 100 to 120 sq ft a litre. The spread for the inside comes from the makers' data sheets; Maker sheets: 0.083 to 0.100 L a m² for two coats.

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Tiles, use by use

Pieces are one over the pitch of the tile in both directions, joint included, on the area to buy.

USE	TILE	NET AREA (M²)	EXTRA	TO BUY (M²)	PIECES
Floors of living rooms, bedrooms, kitchens and passages, with skirting	600 × 600 mm	157.9	5%	165.8	458
Bathroom floors	300 × 300 mm	17.5	10%	19.3	210
Balcony floors	300 × 300 mm	4.5	10%	5.0	54
Bathroom walls, to the door height	300 × 600 mm	55.4	10%	61.0	336
Kitchen wall above the counter	300 × 600 mm	2.5	10%	2.8	16

Floor tile is 600 × 600 mm, laid straight. Extra is for breakage and cutting: 5% straight, 10% diagonal, and 10% for large tiles or small rooms. No standard fixes these; they are Assumed.

Tiles to order

The sizes added up, with the boxes where the maker’s box is known.

TILE	TO BUY (M²)	PIECES	IN A BOX	BOXES
600 × 600 mm	165.8	458	4 pieces, 1.44 m²	115
300 × 300 mm	24.2	264	Not in the data read	–
300 × 600 mm	63.7	351	5 pieces, 0.90 m²	71

Box sizes are from one maker’s data sheet for each size and will differ with the maker. For a tile not in the data read, order by the piece.

Fixing the tiles, grout, skirting and wet areas

A 20 mm bed of 1:4 mortar on floors and 12 mm of 1:3 on walls, with a cement slurry under each (CPWD item text).

ITEM	QUANTITY	UNIT	BASIS
Cement for the bed and the slurry	2,489	kg	50 bags of 50 kg
Sand for the bed	4.56	m³	Coarse sand
Grout	31	kg	Net area of every tile, with 10% extra
Skirting	118.2	m	100 mm high, cut from the floor tile, with 10% for cutting
Waterproofing coat in bathrooms	26.2	m²	Floor and 0.30 m up the walls; at least 2 coats
Corner fillet and glass-fibre mesh	28.8	m	50 × 50 mm fillet; 150 mm mesh over it

Grout is the density of 1.6 kg a litre times the joint length, width and tile thickness (a 600 × 600 mm tile gives 0.10 kg a m²). Skirting runs round the living rooms, bedrooms and passages less the door widths.

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Stone

Thresholds and the kitchen counter.

ITEM	QUANTITY	UNIT	BASIS
Door thresholds	5	nos	4.6 m, one at each bathroom, main and balcony door
Kitchen counter, granite	2.8	m²	2.5 m² with 10% for cutting and the sink; 0.6 m deep
Counter edge, moulded	4.2	m	Front edge of the counter

Doors

Counted from the rooms. Shutters and frames of timber; frames are 100 × 65 mm with a 100 mm horn at each top corner.

DOOR	SIZE (M)	SHUTTER, IS 1003	NUMBER	SHUTTERS (M²)	FRAME TIMBER (M³)	HINGES
Main door	1.00 × 2.10	10 DS 21	1	2.1	0.035	4
Room door	0.90 × 2.10	9 DS 21	4	7.6	0.138	12
Bathroom door	0.80 × 2.10	8 DS 21	3	5.0	0.101	9
Door of the terrace stair room	0.90 × 2.10	9 DS 21	1	1.9	0.034	3
All doors			**9**	**16.6**	**0.309**	**28**

Frame timber is the length of the two sides and the top, with the horns, times the section. The section is Assumed; IS 4021, which fixes frame sections, could not be read.

Windows, ventilators and glazed doors

Sizes are the usual ones; no standard fixes them. Sections are the frame and the sashes, in aluminium or uPVC as you choose.

OPENING	SIZE (M)	NUMBER	GLASS (M²)	SECTIONS (M)
Balcony door, glazed	1.20 × 2.10	1	2.1	17.0
Bedroom window	1.50 × 1.20	3	4.5	38.4
Living room window	1.80 × 1.20	3	5.4	42.0
Kitchen window	1.20 × 1.20	1	1.2	11.6
Stair window	1.20 × 1.20	2	2.4	23.2
Bathroom ventilator	0.60 × 0.45	3	0.7	6.3
All windows and glazed doors		**13**	**16.4**	**138.5**

Glass is 80% of the opening, 4 to 5.5 mm, with 5% for cutting. Sashes are two to a window wider than 0.9 m; a ventilator is a fixed louvre. Grills, nets and the sill are not counted.

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Door hardware

Counted on the rule of one set to a door. Hinges follow IS 1003; the rest is Assumed.

ITEM	NUMBER	BASIS
Door hinges	28	3 for a shutter up to 0.9 m wide, 4 above (IS 1003 cl 8.2)
Lock sets	4	One for the main door and for every bedroom
Tower bolts	5	Two on the main door, one on each bathroom door
Handle pairs	9	One for every door
Door stoppers	9	One for every door

Checked against the Indian Standards

PASS IS 15622:2006 cl 4.1

Tile type for floors and walls

Floor tiles in bathrooms, balconies and kitchens should be of water absorption group BIa (up to 0.08%), which means vitrified or porcelain tiles. Glazed ceramic tiles are for walls. This report counts the sizes you chose; the group is your choice at the shop.

PASS IS 1200 (Part 11):1977, (Part 12):1976, (Part 13):1994; later editions exist (BIS lists 2024 revisions of Parts 11 and 13, not checked)

How the surfaces are measured

Floors are measured before the skirting and dado go on. Openings up to 0.5 m² are not taken out of paint or plaster; from 0.5 to 3 m², one face is taken out. Doors, windows and glazed balcony doors are taken out on that rule, and the skirting is measured in running metres after the door widths. This follows the rules of the editions named, as the tool reads them; the 2024 revisions of Parts 11 and 13 have not been checked.

PASS NBC 2016 Part 3, as summarised in secondary sources; clause numbers not given

Kitchen and bathroom sizes

The rooms in this report are sized from your floor area. The kitchen works out near 9.8 m² against a minimum of 5.0 m² (1.8 m wide), and a bathroom with a WC near 5.8 m² against 2.8 m².

PASS IS 1003 (Part 1):2003 cl 7.2 and Table 3

Door sizes

Doors are sized to the standard designations: 10 DS 21 for the main door (1.0 m), 9 DS 21 for rooms and the kitchen (0.9 m) and 8 DS 21 for bathrooms (0.8 m), all 2.1 m high. The number in the name times 100 mm is the opening width.

PASS NBC 2016 Part 8, as summarised in secondary sources; clause numbers not given

Window size against room size

A habitable room needs windows of at least a tenth of its floor area. The bedroom window counted is 1.8 m² for a room of about 17.7 m² (10%) and the living room has 3.2 m² of window for about 27.2 m² (12%).

NOTE IS 3067:1988 (scope: basements and roofs)

Waterproofing in bathrooms

No Indian Standard read here gives the coverage of a bathroom waterproofing coat; IS 3067 covers basements and roofs. This report counts the area to coat (the floor and 0.30 m up the walls) and the corner fillet, and leaves the coverage to the product's data sheet. Test it by ponding water for 24 hours before the tiles go on.

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PASS IS 2395 (Part 1):1994

Coats of paint

IS 2395 gives no single number of coats for emulsion or distemper, so the coats counted follow the makers' data sheets: cement putty (about 1 mm), one coat of primer, two coats of emulsion.

How far off can this be

- Painted area: from 15% below to 20% above the figure shown.
- Putty, primer and paint: from 30% below to 35% above the figure shown.
- Tiles: from 10% below to 15% above the figure shown.
- Mortar, adhesive and grout: from 35% below to 30% above the figure shown.
- False ceiling: from 15% below to 20% above the figure shown.
- Doors and windows: from 15% below to 25% above the figure shown.
- Plywood and laminate: from 30% below to 40% above the figure shown.

Assumptions

- Code: surfaces are measured following the rules of IS 1200 as the tool reads them, in the editions named under Standards and sources: (Part 11) floors, dado and skirting, (Part 12) plaster and (Part 13) paint. BIS lists 2024 revisions of Parts 11 and 13; the tool has not been checked against them. Openings to 0.5 m² are not taken out, from 0.5 to 3 m² one face is. Door shutters are the standard sizes of IS 1003 (Part 1). Plywood sheets are IS 303.
- Practice: the exterior paint spread is the CPWD Delhi Analysis of Rates 2016 coefficient (cement paint 0.384 kg, smooth acrylic 0.167 L, textured 0.328 L a m²) and the door paint is 0.075 L of primer and 0.090 L of paint a m². The DAR's exterior primer is 0.22 kg a m² in 2016 and 0.09 L in a later schedule; the maker's sheet (0.09 to 0.108 L) agrees with the later figure, so 0.095 L is used.
- Maker: the inside spread (putty, primer, emulsion, distemper) is from the data sheets of Asian Paints, Nerolac and Berger, because the DAR emulsion items could not be read. These vary widely between brands (an emulsion is 0.063 to 0.138 L a m² for two coats), so the range around the paint figure is wide.
- Derived: tiles a m² is one over (length + joint) times (width + joint). Grout is 1.6 kg a litre times the joint length, width and tile thickness. Panel areas for wardrobes and kitchen units are worked out from the box geometry. Board joint and hanger counts for the false ceiling follow the spacing of a public gypsum specification.
- Assumed: the room plan (shared with MasonryPro): a bedroom is one share of the floor, a living room 1.3 shares plus 0.3 for each bedroom, a kitchen 0.75, a bathroom 0.33, and the passages about half a share. Walls are taken as 230 mm outside and 115 mm inside.
- Assumed: clear height 2.85 m. Bathroom wall tiles stop at 2.1 m. The wall above the kitchen counter is tiled 0.6 m high. Kitchen runs are 3 m straight, 4.2 m in an L and 5.4 m in a U; wardrobes are 2.4 m (master) and 1.8 m (others).
- Assumed: wastage of 5% on paint, putty and primer; 5% to 15% on tiles by the layout and the size; 10% on grout; 8% on gypsum board; 12% on plywood and laminate; 10% on granite; 5% on glass. No standard or maker gives these.
- Assumed: door and window sizes and counts: a window in every bedroom and kitchen, two in a living room on the ground floor or in a flat, a ventilator in each bathroom, a stair window for each storey in a building of two floors or more, a glazed door to each balcony, and a stair room on the terrace of any building of two floors or more.
- Assumed: the false ceiling hangers are 0.45 m long, and the cove, if any, is not counted.
- Assumed: the tile bed (mortar or adhesive) covers every tiled area, with the floor bed at 11.0 kg of cement and 0.022 m³ of sand a m² and the wall bed at 9.2 kg and 0.0125 m³ a m². These come from CPWD item text quoted in public tenders; the DAR flooring pages could not be read.
- The plaster underneath, the cement and sand for it, and the walls are in MasonryPro. Putty starts where the plaster ends.

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Not counted in this report

- Plaster, and the cement and sand for it (these are in MasonryPro).
- The flooring screed under tile adhesive, stairs (treads, risers and landings), and the terrace and parapet finish.
- Marble, granite, wood and laminate flooring apart from the thresholds and counter. No source read gives their quantities.
- Sanitary fittings, taps and bathroom accessories (these are in PlumbingPro) and the lights and fans (in ElectricalPro).
- Jointing tape and compound for the false ceiling, cove lighting, access panels, curtain boxes and any bulkhead.
- Drawers, channels, baskets, pull-outs, locks, handles on drawers, mirrors and shelf pins; loft cabinets above 2.1 m.
- Window grills, mosquito nets, sills, curtains and blinds, and timber windows.
- Inside of the terrace stair room, the compound wall and gate, and polishing of timber.
- Scaffolding, masking, putty sanding, curing water and the carriage of materials.

Standards and sources used

- IS 1200 (Part 11):1977, (Part 12):1976, (Part 13):1994 Methods of measurement of building and civil engineering works (BIS lists 2024 revisions of Parts 11 and 13; the tool has not been checked against them)
- IS 15622:2006 Pressed ceramic tiles; IS 2395 (Part 1):1994 Painting practice; IS 3067:1988 Damp-proofing and waterproofing
- IS 1003 (Part 1):2003 Timber panelled and glazed shutters; IS 2202 (Part 2):1983 Flush door shutters; IS 303:2024 Plywood
- CPWD Delhi Analysis of Rates 2016, Volume 2 sub-head 13 (finishing: plaster, paint, primers) and Volume 1 materials list
- Data sheets of paint, putty, adhesive, grout and tile makers (Asian Paints, Nerolac, Berger, Sika, Laticrete, Mapei, Kajaria); public CPWD-pattern tender documents for gypsum ceilings and tile beds

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APPENDIX A

Calculation sheet

InteriorPro: how every figure in this report was worked out

What this sheet is

This sheet shows how InteriorPro turned your answers into the paint, putty, tile, ceiling, door, window and plywood quantities in the report. Each step is written out: the rule, the numbers put in, the result and where the rule comes from, so that a contractor, an architect or a bank's engineer can check the arithmetic. It checks quantities for planning and ordering. It is not a drawing, a bill of quantities or a design.

How to read it

Each row is one step. STEP says what is being found. FORMULA gives it in symbols. WORKING puts the numbers in. RESULT is the answer with its unit. SOURCE names the clause or table the step rests on. Shaded rows are inputs: a figure you gave, or a value read from a code table. Green rows are figures that appear in the report. Every sum was carried out at full precision and is shown rounded, so re-keying the rounded numbers by hand may differ in the last digit.

Method in brief

1. Your answers are read and the clear height of a room is found: the floor to floor height less the slab (section 1).
2. Each floor is shared out among its rooms in proportion to a weight for each kind of room, and the length of wall round each room follows from its shape. This room plan is an assumption and is the same one MasonryPro uses (section 2).
3. Doors, windows, ventilators and balcony doors are counted from the rooms by fixed rules (section 3).
4. The surfaces are measured following the rules of IS 1200 as the tool reads them (editions named in the references; the 2024 revisions have not been checked): walls at clear height with openings taken out by the 0.5 m² and 3 m² rules, ceilings, the outside from the plinth to the parapet, and doors on both faces (section 4).
5. Tiles are found use by use: net area, an allowance for breakage, pieces from the size of the tile plus its joint, then boxes. Skirting, the mortar bed or the adhesive, and the grout follow (section 5).
6. Bathroom waterproofing, door thresholds and the kitchen counter are measured from the same rooms (section 6).
7. Putty, primer and paint are each the area to be painted times a spread from a maker or from the CPWD analysis of rates, times the wastage (section 7).
8. No false ceiling was chosen, so none is counted.
9. Door shutters, frames, glass, window sections and hardware follow from the counts (section 8). No wardrobes or kitchen units were chosen, so no plywood or laminate is counted.
10. Every figure is rounded up to something that can be bought (section 9). The room and window sizes quoted in the report are shown (section 10), and the likely range of each total is applied (section 11).
11. Every total, table figure and buy-list line of the report is rebuilt here and compared with the report's own in the reconciliation. Had one differed, this sheet would have been left out of the report.

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Read this before relying on the sheet

- The rooms are not measured. Their sizes come from a room plan that shares each floor out by fixed weights, so the wall, floor and ceiling areas, and the number of doors and windows, will differ from your drawings. The ranges in section 11 say how far.
- Each NirmanShastra tool lays out the same house by its own rule, so the area of a room in this report can differ noticeably from the same room in another NirmanShastra report. Only MasonryPro uses the same room plan as this report.
- The surfaces follow the rules of IS 1200 Part 11 (1977 edition) and Part 13 (1994 edition) as the tool reads them. BIS lists 2024 revisions of Parts 11 and 13, and the tool has not been checked against them. No clause number of IS 1200 is quoted here. One difference is already suspected: a summary of the BIS draft of the 2024 revision of Part 13 suggests that, where the two faces of a wall have different finishes, an opening between 0.5 and 3 m² is taken out first from the face with the narrower reveal and is halved only where the reveals are equal. This tool halves it for every window and outer door. The total area taken out is the same either way; only the split between inside and outside paint would move. This has not been checked against the full text.
- The spreads of paint, putty, adhesive and grout, the tile boxes and the counts for a false ceiling come from makers' data sheets, the CPWD Delhi Analysis of Rates 2016 and a public tender specification. They are not Indian Standards, they differ from brand to brand and from site to site, and none was tested on a site.
- The allowances for wastage are assumptions: paint 5%, tiles 5% to 15%, grout 10%, gypsum board 8%, plywood and laminate 12%, granite 10%, glass 5%. No standard or maker gives them. Frame timber has no allowance for converting rough timber (sawing, planing and waste).
- Door and window sizes, counts and frame sections are typical values. The frame section could not be checked against IS 4021.
- The shutter area of a door is taken as the area of the whole opening. For 9 DS 21 the shutter of IS 1003 (Part 1) Table 3 is 1.60 m² and the opening is 1.89 m², so the area used is about 18% more than the shutter, and the door paint in section 7 is worked on that larger area. Hinges are counted from the width of the opening (three up to 0.9 m, four above), but IS 1003 (Part 1) cl. 8.2 speaks of the width of the shutter: a 900 mm shutter takes three hinges, and the main door (1.00 m opening, 10 DS 21) is given 4 here, one more than the standard would give.
- Nothing is designed here. Waterproofing is counted by area only (IS 3067 covers basements and roofs, so the coverage is left to the product's data sheet). The screed under tile adhesive, stairs, plaster, polishing and the terrace finish are not counted.
- The kitchen and bathroom size checks and the window size check quoted in the report are NBC 2016 figures as summarised in secondary sources. The Part and clause numbers were not read in the code, and no clause number is given. They are advisory and change no quantity.
- A clause or item number given here for IS 4021, IS 3067, IS 303 or a CPWD analysis-of-rates item is a pointer to check, not a quotation. Such numbers are as given in the tool's research notes and were not re-read against the published text. The clause numbers of IS 1003 (Part 1) Table 3 and cl. 8.2 were read.
- Hardware is one set to a door; glazed balcony doors are given none. Drawers, channels, cabinet locks, grills and nets are not counted.
- The sheet is quantities only: no rates, no labour, no cost. The state and city do not change any quantity. Figures are carried at full precision and shown rounded, so re-keying the printed numbers can differ in the last digit.

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1. Your answers and the building

Your answers

QUESTION	YOUR ANSWER
Building	A house
Floors	Ground floor and 1 above
Built-up area of each floor	1,000, 1,000 sq ft
Bedrooms in the house	3
Bathrooms in the house	3
Kitchens in the house	1
Balconies in the house	1
Floor to floor height	3.00 m
Paint inside	Standard: putty, primer and emulsion
Paint outside	Smooth acrylic exterior emulsion, 2 coats over exterior primer
False ceiling	None
Floor tile	600 × 600 mm, laid straight, fixed with a mortar bed
Bathroom wall tiles	To the door height
Kitchen	L shape, counter only
Built-in wardrobes	None
State and city	Lucknow, Uttar Pradesh

Heights and units

STEP	FORMULA	WORKING	RESULT	SOURCE
Floor to floor height			**3.00 m**	Your answer
Thickness of a floor slab Taken for every floor and for the terrace slab			**0.15 m**	NirmanShastra assumption
Clear height of a room Floor to the underside of the slab above. Wall paint and plaster use this height.	Hc = H − slab	3.00 − 0.15	**2.85 m**	Engine rule
Square feet to square metres			**0.09290304 m² per sq ft**	Definition: 1 ft = 0.3048 m, so 1 sq ft = 0.3048 × 0.3048 m²

Floor areas and rooms on each floor

Area in m² = area in sq ft × 0.09290304.

FLOOR	BUILT-UP (SQ FT)	BUILT-UP (M²)	HOMES	BEDROOMS	BATHROOMS	KITCHENS	BALCONIES ON THE FLOOR
Ground floor	1,000	92.90	1	2	2	1	0
First floor	1,000	92.90	1	1	1	0	1

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2. The room plan

Rooms are not measured from a drawing. The plan below is what the tool assumes, and everything after it follows from it.

Room plan, worked for the ground floor (92.9 m²)

No drawing is read. Each floor is shared out among its rooms in proportion to a weight for each kind of room. These weights are the tool's own assumption, the same ones MasonryPro uses, so both reports measure the same walls.

STEP	FORMULA	WORKING	RESULT	SOURCE
Outer wall, all round the floor The wall round a plan whose sides are in the ratio 1 to 1.5 is 2 × (1 + 1.5) ÷ √1.5 = 4.0825 √A. InteriorPro uses 4.082.	P = 4.082 × √A	4.082 × √92.90	**39.34 m**	NirmanShastra assumption (room plan shared with MasonryPro)
Rooms in one home on this floor			**2 bedrooms, 2 bathrooms, 1 kitchen, 1 living room**	NirmanShastra assumption (room plan shared with MasonryPro)
Weight of the living room Ground floor and flats. On the upper floors of a house the weight is 0.9 + 0.2 × bedrooms.	wL = 1.3 + 0.3 × bedrooms	1.3 + 0.3 × 2	**1.90**	NirmanShastra assumption (room plan shared with MasonryPro)
Weight of the passages	wP = 0.45 + 0.08 × (bedrooms + bathrooms + kitchens)	0.45 + 0.08 × (2 + 2 + 1)	**0.85**	NirmanShastra assumption (room plan shared with MasonryPro)
Total weight of one home	Σw = bedrooms × 1.0 + wL + kitchens × 0.75 + bathrooms × 0.33 + wP	2 × 1 + 1.90 + 1 × 0.75 + 2 × 0.33 + 0.85	**6.160**	NirmanShastra assumption (room plan shared with MasonryPro)
Inner walls, as found by the plan Inner wall = (the wall round all the rooms − the outer wall) ÷ 2, less the walls between flats. Room sizes depend on it and it depends on room sizes, so the plan repeats the working four times, starting from a floor area of 0.88 A.			**28.03 m**	NirmanShastra assumption (room plan shared with MasonryPro)
Floor area inside the walls Walls are taken as 230 mm outside and 115 mm inside.	net = A − P × 0.23 − Lint × 0.115 − Lp × 0.23 (not less than 0.6 A)	A − 39.34 × 0.23 − 28.03 × 0.115 − 0.00 × 0.23	**80.63 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
Floor area for one unit of weight	u = (net − lobby) ÷ homes ÷ Σw	(80.63 − 0.00) ÷ 1 ÷ 6.160	**13.089 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
One bedroom	a = u × 1.0	13.089 × 1	**13.09 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
The living room	a = u × wL	13.089 × 1.90	**24.87 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
One kitchen	a = u × 0.75	13.089 × 0.75	**9.82 m²**	NirmanShastra assumption (room plan shared with MasonryPro)

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One bathroom	a = u × 0.33	13.089 × 0.33	**4.32 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
The passages, all together	a = u × wP	13.089 × 0.85	**11.13 m²**	NirmanShastra assumption (room plan shared with MasonryPro)

Room plan, every floor

FLOOR	OUTER WALL (M)	INSIDE AREA (M²)	BEDROOM (M²)	LIVING ROOM (M²)	KITCHEN (M²)	BATHROOM (M²)	PASSAGES (M²)	LOBBY (M²)
Ground floor	39.34	80.63	13.09	24.87	9.82	4.32	11.13	–
First floor	39.34	81.79	26.91	29.60	20.18	8.88	16.41	–

One bedroom, measured as an example (ground floor)

The report adds up every room in the building the same way, so this one room shows the whole method. The totals below are what the report uses.

STEP	FORMULA	WORKING	RESULT	SOURCE
Floor area of the bedroom From the room plan above			**13.09 m²**	NirmanShastra assumption (room plan shared with MasonryPro)
Wall round the bedroom Each room is taken as a rectangle whose long side is 1.3 times the short side.	p = 2 (1 + r) ÷ √r × √a	2 × (1 + 1.3) ÷ √1.3 × √13.09	**14.60 m**	NirmanShastra assumption (room plan shared with MasonryPro)
Wall, floor to slab	p × Hc	14.596 × 2.850	**41.60 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Door taken out (once) A door of 0.5 to 3 m² between two painted rooms is taken out once.	width × height	0.9 × 2.1	**1.89 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Window taken out (half) Half comes off the inside and half off the outside, because the finish differs.	0.5 × width × height	0.5 × 1.5 × 1.2	**0.90 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Wall to be painted	wall – door – window	41.60 – 1.89 – 0.90	**38.81 m²**	
Ceiling to be painted	ceiling = floor area	13.09	**13.09 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Floor to be tiled Measured before the skirting goes on.	floor = floor area	13.09	**13.09 m²**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)
Skirting round the bedroom Taken out twice because the skirting is broken on both sides of the door (the bedroom and the room it opens into).	p – 2 door widths	14.60 – 2 × 0.9	**12.80 m**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)
Tile for the skirting The skirting is cut from the floor tile: 100 mm high, 10% extra for cutting.	length × height × 1.10	12.80 × 0.10 × 1.10	**1.408 m²**	Engine rule

Rooms and their walls, every floor

FLOOR	ROOM	NUMBER	AREA OF ONE (M²)	WALL ROUND ONE (M)	TOTAL AREA (M²)	TOTAL WALL (M)
Ground floor	Bedrooms	2	13.09	14.60	26.18	29.19
Ground floor	Living rooms	1	24.87	20.12	24.87	20.12

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Ground floor	Kitchens	1	9.82	12.64	9.82	12.64
Ground floor	Bathrooms	2	4.32	8.38	8.64	16.77
Ground floor	Passages	1	11.13	16.68	11.13	16.68
First floor	Bedrooms	1	26.91	20.93	26.91	20.93
First floor	Living rooms	1	29.60	21.95	29.60	21.95
First floor	Kitchens	0	20.18	18.12	0.00	0.00
First floor	Bathrooms	1	8.88	12.02	8.88	12.02
First floor	Passages	1	16.41	20.26	16.41	20.26

Room totals, whole building

ROOM	NUMBER	FLOOR AREA (M²)	WALL LENGTH (M)
Bedrooms	3	53.08	50.12
Living rooms	2	54.47	42.07
Kitchens	1	9.82	12.64
Bathrooms	3	17.52	28.79
Passages	–	27.54	36.93
Stair lobbies	–	0.00	0.00

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3. Openings

Doors and windows counted from the rooms

OPENING	SIZE (M)	AREA OF ONE (M²)	HOW MANY ARE COUNTED	NUMBER
Main door	1.00 × 2.10	2.10	One, on the ground floor	1
Room door	0.90 × 2.10	1.89	One to each bedroom and each kitchen	4
Bathroom door	0.80 × 2.10	1.68	One to each bathroom	3
Door of the terrace stair room	0.90 × 2.10	1.89	One, if the building has two floors or more	1
Balcony door, glazed	1.20 × 2.10	2.52	One to each balcony	1
Bedroom window	1.50 × 1.20	1.80	One to each bedroom	3
Living room window	1.80 × 1.20	2.16	Two to a living room on the ground floor or in a flat, one on the upper floors of a house	3
Kitchen window	1.20 × 1.20	1.44	One to each kitchen	1
Stair window	1.20 × 1.20	1.44	One on each floor, if the building has two floors or more	2
Bathroom ventilator	0.60 × 0.45	0.27	One to each bathroom	3

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4. Surfaces measured

Measured following the rules of IS 1200 as the tool reads them (editions named in the references; the 2024 revisions have not been checked): openings up to 0.5 m² are not taken out; from 0.5 m² to 3 m², one face is taken out (a door between two painted rooms once, a window or an outer door half from each face because the finish differs). Plaster is not here: it belongs to MasonryPro.

Walls of the rooms and passages

STEP	FORMULA	WORKING	RESULT	SOURCE
Walls, before openings Every room is measured round its own walls, so a wall between two rooms is counted once for each face, as it is painted on both. Bathrooms are done separately.	Walls = (bedrooms + living rooms + kitchens + passages + lobby wall) × Hc	(50.12 + 42.07 + 12.64 + 36.93 + 0.00) × 2.850	**404.0 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Room doors taken out Each is 0.5 to 3 m² and has the same finish on both faces, so one face is taken out.	doors × width × height	4 × (0.9 × 2.1)	**7.56 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Main doors taken out The finish differs inside and outside, so half comes off each face.	doors × width × height × share	1 × (1 × 2.1) × 0.5	**1.05 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Windows and balcony doors taken out (inside half) Different finishes inside and outside, equal reveals: half of the opening comes off each face.	0.5 × Σ number × area	0.5 × (3 × 1.80 + 3 × 2.16 + 1 × 1.44 + 2 × 1.44 + 1 × 2.52)	**9.36 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Tiled wall above the kitchen counter Counter run and 0.6 m of tile above it; this wall is tiled, not painted.	kitchens × run × height	1 × 4.2 × 0.6	**2.52 m²**	NirmanShastra assumption
Total taken out of the walls	doors + windows + tiled wall	7.56 + 1.05 + 9.36 + 2.52	**20.49 m²**	
Walls to be painted	gross − taken out (not below zero)	larger of 0 and (404.0 − 20.5)	**383.5 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)

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Bathroom walls: the tiled part and the painted part

STEP	FORMULA	WORKING	RESULT	SOURCE
Height of the bathroom wall tiling To the door height (2.1 m)			**2.1 m**	NirmanShastra assumption
Bathroom wall tiles A dado is measured in square metres.	wall length × tiling height − door × width × lower of door height and tiling height	larger of 0 and (28.8 × 2.1 − 3 × 0.8 × lower of 2.1 and 2.1)	**55.4 m²**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)
Bathroom wall above the tiles	wall length × (Hc − tiling height)	28.8 × (2.85 − 2.1)	**21.6 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Door taken out above the tiles	doors × width × (door height − tiling height, not below zero)	3 × 0.8 × larger of 0 and (2.1 − 2.1)	**0.0 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Bathroom walls to be painted	above − door	larger of 0 and (21.6 − 0.0)	**21.6 m²**	

Ceilings and the whole inside

STEP	FORMULA	WORKING	RESULT	SOURCE
All inside walls to be painted	rooms + bathrooms	383.5 + 21.6	**405.1 m²**	
Ceilings Painted whether the ceiling is the slab or a false ceiling board. Nothing is taken out.	sum of every room floor area	53.1 + 54.5 + 9.8 + 17.5 + 27.5 + 0.0	**162.4 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Inside area to be painted	walls + ceilings	405.13 + 162.43	**567.6 m²**	

Outside of the building

Measured from the plinth to the top of the parapet.

STEP	FORMULA	WORKING	RESULT	SOURCE
Outer face of the walls The plinth shows 0.45 m above the ground. The stair room on the terrace has 12 m of wall, 2.4 m high plus the slab (0.15 m).	Σ (outer wall × H) + plinth + stair room	236.07 + 39.34 × 0.45 + 30.6	**284.4 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)
Parapet, both faces and top A parapet 1.0 m high and 115 mm thick.	top outer wall × (2 × height + thickness)	39.34 × (2 × 1 + 0.115)	**83.2 m²**	NirmanShastra assumption
Balcony edges and railing walls Each balcony is taken as 6.0 m of edge with a 1.0 m wall.	balconies × edge × (2 × height + thickness)	1 × 6 × (2 × 1 + 0.115)	**12.7 m²**	NirmanShastra assumption
Underside of the balconies Each balcony is taken as 4.5 m².	balconies × area	1 × 4.5	**4.5 m²**	NirmanShastra assumption
Windows and outer doors taken out (outside half) The other half was taken off the inside. A flat main door opens to the lobby and is not taken from the outside.	0.5 × Σ number × area, with the main door of a house	0.5 × (3 × 1.80 + 3 × 2.16 + 1 × 1.44 + 2 × 1.44 + 1 × 2.52 + 1 × 2.10)	**10.41 m²**	IS 1200 (Part 13):1994, rules for deduction of openings (a 2024 revision exists; not checked)
Outside, before openings	faces + parapet + balcony edges + balcony undersides	284.4 + 83.2 + 12.7 + 4.5	**384.8 m²**	
Outside to be painted	gross − taken out (not below zero)	larger of 0 and (384.8 − 10.41)	**374.4 m²**	IS 1200 (Part 13):1994 (a 2024 revision exists; not checked)

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FLOOR	OUTER WALL (M)	FLOOR TO FLOOR (M)	OUTER FACE (M²)
Ground floor	39.34	3.00	118.03
First floor	39.34	3.00	118.03
All floors			**236.07**

Doors, both faces

STEP	FORMULA	WORKING	RESULT	SOURCE
Doors counted	main + room + bathroom + stair room	1 + 4 + 3 + 1	**9**	
Door shutters, one face The size of the opening is used for the shutter.	Σ number × width × height	1 × 2.10 + 4 × 1.89 + 3 × 1.68 + 1 × 1.89	**16.59 m²**	IS 1003 (Part 1):2003 Table 3 (designations)
Doors to be painted, both faces Frames are painted with the shutters and are not measured separately.	2 × shutters	2 × 16.59	**33.2 m²**	Engine rule

Floors to be tiled

STEP	FORMULA	WORKING	RESULT	SOURCE
Floors of living rooms, bedrooms, kitchens and passages Measured before the skirting and dado go on. Bathroom floors and balconies are tiled in a smaller tile and are counted below.	sum of the floor areas	53.1 + 54.5 + 9.8 + 27.5 + 0.0	**144.9 m²**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)
Balcony floors	balconies × area	1 × 4.5	**4.5 m²**	NirmanShastra assumption
Floors to be tiled, in all	rooms + bathrooms + balconies	144.9 + 17.5 + 4.5	**166.9 m²**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)

The surfaces, as listed in the report

These are the results of the steps above, set out as in the report table "Surfaces, measured".

SURFACE	MEASURED (M²)	TAKEN OUT (M²)	NET (M²)
Walls in rooms and passages	404.0	20.5	383.5
Bathroom walls above the tiles	21.6	0.0	21.6
Ceilings, inside	162.4	0.0	162.4
Outside of the building	384.8	10.4	374.4
Doors, both faces	33.2	0.0	33.2
Bathroom wall tiles	60.5	5.0	55.4
Kitchen wall above the counter, tiles	2.5	0.0	2.5
Floors to be tiled	166.9	0.0	166.9

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5. Floor and wall tiles, skirting, fixing and grout

Floors are measured before the skirting and dado go on (IS 1200 Part 11, 1977 edition; a 2024 revision exists and has not been checked). Pieces are counted from the size of the tile plus its joint.

Pieces to the square metre

STEP	FORMULA	WORKING	RESULT	SOURCE
Tiles in one square metre (floor tile) Lengths in metres. Each tile takes its own size plus one joint in each direction.	n = 1 ÷ ((L + J) × (W + J))	1 ÷ ((0.6 + 0.002) × (0.6 + 0.002))	**2.759 tiles**	Derived: arithmetic shown

Sizes of the tiles used

TILE	LENGTH (MM)	WIDTH (MM)	JOINT (MM)	THICKNESS (MM)	TILES IN 1 M²
600 × 600 mm	600	600	2.0	9.0	2.76
300 × 300 mm	300	300	3.0	8.0	10.89
300 × 600 mm	300	600	2.0	8.0	5.50

Skirting

STEP	FORMULA	WORKING	RESULT	SOURCE
Skirting, running metres Skirting is measured in running metres. Each bedroom door is taken out twice (the skirting is broken on both sides of it), each kitchen, bathroom, main and balcony door once. Kitchen and bathroom walls are tiled to a height, so they carry no skirting.	wall round bedrooms, living rooms, passages and lobby – door widths	larger of 0 and (50.1 + 42.1 + 36.9 + 0.0 – (2 × 3 × 0.9 + 1 × 0.9 + 3 × 0.8 + 1 × 1 + 1 × 1.2))	**118.2 m**	IS 1200 (Part 11):1977 (a 2024 revision exists; not checked)
Tile for the skirting Skirting is 100 mm high and cut from the floor tile; 10% is allowed for cutting.	length × height × (1 + cutting)	118.2 × 0.1 × 1.1	**13.00 m²**	NirmanShastra assumption

Tile area and pieces, use by use

The first use is worked in full; the rest follow the same steps and are in the table.

STEP	FORMULA	WORKING	RESULT	SOURCE
Floor tile, net area	floors of rooms + skirting	144.9 + 13.00	**157.9 m²**	
Allowance for breakage and cutting No standard fixes it. Straight laying 5%, diagonal 10%; large tiles (800 × 800, 600 × 1200) 5 points more; 10% in bathrooms, on balconies and for the kitchen wall.	by layout, tile size and room size		**5%**	NirmanShastra assumption
Floor tile to buy	net × (1 + allowance)	157.9 × (1 + 0.05)	**165.8 m²**	
Floor tile, pieces	pieces = whole number above (area to buy × tiles in 1 m²)	next whole number above 165.8 × 2.759	**458 tiles**	Derived: arithmetic shown

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USE	TILE	NET AREA (M²)	EXTRA	TO BUY (M²)	PIECES
Floors of rooms, with skirting	600 × 600 mm	157.9	5%	165.8	458
Bathroom floors	300 × 300 mm	17.5	10%	19.3	210
Balcony floors	300 × 300 mm	4.5	10%	5.0	54
Bathroom walls	300 × 600 mm	55.4	10%	61.0	336
Kitchen wall above the counter	300 × 600 mm	2.5	10%	2.8	16

Tiles to order, by size

STEP	FORMULA	WORKING	RESULT	SOURCE
Tiles to buy, all sizes	sum of the area to buy of each size	165.81 + 24.22 + 63.74	**253.8 m²**	

TILE	NET (M²)	TO BUY (M²)	PIECES	IN A BOX	BOXES
600 × 600 mm	157.9	165.8	458	4 pieces, 1.44 m²	115
300 × 300 mm	22.0	24.2	264	Not in the data read	–
300 × 600 mm	57.9	63.7	351	5 pieces, 0.90 m²	71

Fixing: mortar bed and slurry

STEP	FORMULA	WORKING	RESULT	SOURCE
Area fixed on the floor bed	rooms + bathrooms + balconies	144.9 + 17.5 + 4.5	**166.9 m²**	
Area fixed on the wall bed	bathroom walls + kitchen wall + skirting	55.4 + 2.5 + 13.00	**70.9 m²**	
Cement for the bed and slurry The kilograms of cement in the bed and slurry for one m².	floor × 11.0 + wall × 9.2	166.93 × 11 + 70.95 × 9.2	**2,489 kg**	CPWD item text as quoted in public tenders (20 mm bed of 1:4 mortar on floors, 12 mm of 1:3 on walls, with a cement slurry); cement worked from the DAR yield
Sand for the bed	floor × 0.022 + wall × 0.0125	166.93 × 0.022 + 70.95 × 0.0125	**4.56 m³**	CPWD item text as quoted in public tenders
Bags of cement	whole bags of 50 kg	next whole number above 2,489 ÷ 50	**50 bags**	

Grout

STEP	FORMULA	WORKING	RESULT	SOURCE
Grout in 1 m² of 600 × 600 mm tiling Joint length in 1 m² × joint width × tile thickness gives the volume of grout; 1.6 kg a litre is the density InteriorPro assumes (1.4 to 1.6).	g = 1.6 × 1000 × (1 ÷ L + 1 ÷ W) × J × T	1.6 × 1000 × (1 ÷ 0.6 + 1 ÷ 0.6) × 0.002 × 0.009	**0.096 kg**	Derived: arithmetic shown
Grout, all sizes Net area of every tile (no breakage), with 10% extra for wastage. The 10% is an assumption.	Σ net area × kg a m² × 1.10	16.7 + 6.2 + 8.2	**31 kg**	Derived: arithmetic shown

TILE	NET AREA (M²)	GROUT (KG A M²)	GROUT WITH 10% EXTRA (KG)
600 × 600 mm	157.9	0.096	16.7
300 × 300 mm	22.0	0.256	6.2
300 × 600 mm	57.9	0.128	8.2

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6. Wet areas and stone

Bathroom waterproofing, thresholds and the kitchen counter

STEP	FORMULA	WORKING	RESULT	SOURCE
Waterproofing coat in the bathrooms The floor and 0.30 m up the walls. Test by ponding water for 24 hours before tiling.	bathroom floors + wall length × 0.30	17.52 + 28.79 × 0.3	**26.2 m²**	IS 3067:1988 (area only; its scope is basements and roofs, so the coverage is left to the product's data sheet)
Corner fillet and mesh	wall length of the bathrooms	28.8	**28.8 m**	NirmanShastra assumption
Door thresholds, length	bathroom + main + balcony door widths	3 × 0.8 + 1 × 1 + 1 × 1.2	**4.6 m**	NirmanShastra assumption
Door thresholds, number	bathroom + main + balcony doors	3 + 1 + 1	**5**	
Kitchen counter, granite Counter 0.60 m deep along the kitchen run you chose.	kitchens × run × depth	1 × 4.2 × 0.6	**2.52 m²**	NirmanShastra assumption
Granite to buy 10% for cutting and the sink.	counter × (1 + cutting)	2.52 × 1.1	**2.8 m²**	NirmanShastra assumption
Counter edge, moulded	kitchens × run	1 × 4.2	**4.2 m**	NirmanShastra assumption

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7. Putty, primer and paint

Putty, primer and paint are counted by the area they cover. Paint area × spread × 1.05 gives what to buy. The number of coats is already inside each spread.

Painted area

STEP	FORMULA	WORKING	RESULT	SOURCE
Painted area, inside and outside Doors are measured on their own and are not in this total.	inside + outside	567.55 + 374.37	**941.9 m²**	
Wastage on putty, primer and paint No maker or CPWD sheet gives a wastage figure.	quantity = area × spread × (1 + wastage)		**5%**	NirmanShastra assumption

How far each layer spreads, and the coats counted

LAYER	SPREAD, ALL COATS, A M²	COVERS (M² PER KG OR L)	COVERS (SQ FT PER KG OR L)	COATS	SOURCE
Wall putty, cement based	0.600 kg	1.7	18	about 1 mm, in coats as the maker advises	Maker sheet: 10 to 15 sq ft a kg for 1.5 mm in two coats, scaled to 1 mm (0.48 to 0.72 kg a m²)
Primer, inside	0.075 L	13.3	144	one coat	Maker sheet: 130 to 160 sq ft a litre (0.067 to 0.083 L a m²)
Interior emulsion, 2 coats	0.090 L	11.1	120	Cement putty (about 1 mm), one coat of primer, two coats of emulsion	Maker sheets: 0.083 to 0.100 L a m² for two coats
Primer, outside	0.095 L	10.5	113	one coat	Maker sheet, 100 to 120 sq ft a litre; the DAR figure differs (see the report assumptions)
Exterior acrylic emulsion	0.167 L	6.0	64	Smooth acrylic exterior emulsion, 2 coats over exterior primer	CPWD DAR 2016 item 13.46.1 (0.167 L a m²); primer from the maker sheet, 100 to 120 sq ft a litre
Primer, doors	0.075 L	13.3	144	one coat	CPWD DAR 2016 item 13.48.2
Enamel or multi-surface paint, doors	0.090 L	11.1	120	two coats	CPWD DAR 2016 item 13.50.1

Putty, primer and paint to order

STEP	FORMULA	WORKING	RESULT	SOURCE
Wall putty, cement based Inside walls and ceilings	quantity = area × spread × (1 + wastage)	567.6 × 0.6 × 1.05	**357.6 kg**	Makers' data sheets / CPWD DAR 2016 (spread table above)
Primer, inside Inside walls and ceilings	quantity = area × spread × (1 + wastage)	567.55 × 0.075 × 1.05	**44.69 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)
Interior emulsion, 2 coats Inside walls and ceilings	quantity = area × spread × (1 + wastage)	567.55 × 0.09 × 1.05	**53.63 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)

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Primer, outside Outside walls	quantity = area × spread × (1 + wastage)	374.37 × 0.095 × 1.05	**37.34 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)
Exterior acrylic emulsion Outside walls	quantity = area × spread × (1 + wastage)	374.4 × 0.167 × 1.05	**65.65 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)
Primer, doors Doors, both faces	quantity = area × spread × (1 + wastage)	33.2 × 0.075 × 1.05	**2.61 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)
Enamel or multi-surface paint, doors Doors, both faces	quantity = area × spread × (1 + wastage)	33.2 × 0.09 × 1.05	**3.14 L**	Makers' data sheets / CPWD DAR 2016 (spread table above)

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8. Doors, windows, frames and hardware

Sizes and counts are assumptions; the door designations are those of IS 1003 (Part 1). The shutter area is that of the whole opening, and the hinges follow the width of the opening (see the notes under the door table).

Doors and their frames

STEP	FORMULA	WORKING	RESULT	SOURCE
Timber frame of one room door Two sides and the head, with a 100 mm horn at each top corner, in a section of 100 × 65 mm. IS 4021, which fixes frame sections, could not be read, so the section is an assumption. This is the net volume of the frame: no allowance is added for converting rough timber (sawing, planing and waste).	V = (2 × height + width + 2 × horn) × section width × section depth	(2 × 2.1 + 0.9 + 2 × 0.1) × 0.1 × 0.065	**0.0345 m³**	NirmanShastra assumption
Hinges on one room door Applied here to the width of the opening; the standard speaks of the width of the shutter. A 900 mm shutter takes three hinges, so the main door (1.00 m opening, 10 DS 21) is given 4 here, one more than the standard would give its 900 mm shutter. The report keeps the count of 4.	three up to 0.9 m wide, four above	0.9 m wide, so 3	**3**	IS 1003 (Part 1):2003 cl. 8.2

DOOR	SIZE (M)	SHUTTER	NUMBER	SHUTTERS (M²)	FRAME TIMBER (M³)	HINGES ON ONE	HINGES
Main door	1.00 × 2.10	10 DS 21	1	2.1	0.035	4	4
Room door	0.90 × 2.10	9 DS 21	4	7.6	0.138	3	12
Bathroom door	0.80 × 2.10	8 DS 21	3	5.0	0.101	3	9
Door of the terrace stair room	0.90 × 2.10	9 DS 21	1	1.9	0.034	3	3
All doors			**9**	**16.6**	**0.309**		**28**

Windows, ventilators and glazed doors

STEP	FORMULA	WORKING	RESULT	SOURCE
Glass in one bedroom window Glass is 80% of the opening, in 4 to 5.5 mm, with 5% for cutting.	area × glass share × (1 + cutting)	1.5 × 1.2 × 0.8 × 1.05	**1.512 m²**	NirmanShastra assumption
Outer frame of the window	2 × (width + height)	2 × (1.5 + 1.2)	**5.40 m**	NirmanShastra assumption
Sashes of the window One pane up to 0.9 m wide, two above. A ventilator is a fixed louvre with no sash.	panes × 2 × (height − 0.1 + width ÷ panes)	2 × 2 × (1.2 − 0.1 + 1.5 ÷ 2)	**7.40 m**	NirmanShastra assumption
Aluminium or uPVC sections in one bedroom window	frame + sashes	5.40 + 7.40	**12.80 m**	

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OPENING	SIZE (M)	NUMBER	GLASS (M²)	SECTIONS (M)
Balcony door, glazed	1.20 × 2.10	1	2.1	17.0
Bedroom window	1.50 × 1.20	3	4.5	38.4
Living room window	1.80 × 1.20	3	5.4	42.0
Kitchen window	1.20 × 1.20	1	1.2	11.6
Stair window	1.20 × 1.20	2	2.4	23.2
Bathroom ventilator	0.60 × 0.45	3	0.7	6.3
All windows and glazed doors		**13**	**16.4**	**138.5**

Door hardware

Hinges are the total of the door table. The rest follow one set to a door.

STEP	FORMULA	WORKING	RESULT	SOURCE
Lock sets One for the main door and one for every bedroom	main doors + bedrooms	1 + 3	**4**	NirmanShastra assumption
Tower bolts	2 × main doors + bathroom doors	2 × 1 + 3	**5**	NirmanShastra assumption
Handle pairs Glazed balcony doors are not given hardware in this count.	one for every door	9	**9**	NirmanShastra assumption
Door stoppers	one for every door	9	**9**	NirmanShastra assumption

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9. The buy list

What to order, and how each figure was rounded

ITEM	WORKED OUT AS	ROUNDED	TO ORDER	UNIT
Wall putty, cement based	357.6 kg	Up to the next multiple of 5 kg	360	kg
Primer, inside	44.69 L	Next whole number above	45	L
Interior emulsion, 2 coats	53.63 L	Next whole number above	54	L
Primer, outside	37.34 L	Next whole number above	38	L
Exterior acrylic emulsion	65.65 L	Next whole number above	66	L
Primer for doors	2.61 L	Next whole number above	3	L
Enamel for doors	3.14 L	Next whole number above	4	L
Tile 600 × 600 mm	165.8 m²	Pieces worked out in section 5	458	nos
Tile 300 × 300 mm	24.2 m²	Pieces worked out in section 5	264	nos
Tile 300 × 600 mm	63.7 m²	Pieces worked out in section 5	351	nos
Cement	see section 5	Whole bags of 50 kg	50	bags of 50 kg
Sand	4.56 m³	Rounded to 0.1 m³	4.6	m³
Tile grout	31.0 kg	Next whole number above	32	kg
Waterproofing coat, bathrooms	26.16 m²	Rounded to 0.1 m²	26.2	m²
Kitchen counter, granite	2.77 m²	Rounded to 0.1 m²	2.8	m²
Counter edge, moulded	4.20 m	Rounded to 0.1 m	4.2	m
Door thresholds	4.60 m	Rounded to 0.1 m	4.6	m
Main door, 10 DS 21	counted in section 3	Whole doors	1	nos
Room door, 9 DS 21	counted in section 3	Whole doors	4	nos
Bathroom door, 8 DS 21	counted in section 3	Whole doors	3	nos
Door of the terrace stair room, 9 DS 21	counted in section 3	Whole doors	1	nos
Door frame timber	0.309 m³	Rounded to 0.01 m³	0.31	m³
Door hinges	see section 8	Whole numbers	28	nos
Lock sets	see section 8	Whole numbers	4	nos
Tower bolts	see section 8	Whole numbers	5	nos
Handle pairs	see section 8	One pair to a door	9	pairs
Door stoppers	see section 8	One to a door	9	nos
Window glass, 4 to 5.5 mm	16.41 m²	Rounded to 0.1 m²	16.4	m²
Window and sash sections	138.50 m	Next whole number above	139	m

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10. Room and window sizes quoted in the report

The report checks the kitchen, the bathroom and the windows against usual minimums. The sizes it quotes are worked out here.

Room sizes and window sizes quoted in the report

STEP	FORMULA	WORKING	RESULT	SOURCE
Average kitchen	kitchen floor area ÷ number of kitchens	9.82 ÷ 1	**9.8 m²**	Derived: arithmetic shown
Average bathroom	bathroom floor area ÷ number of bathrooms	17.52 ÷ 3	**5.8 m²**	Derived: arithmetic shown
Average bedroom	bedroom floor area ÷ number of bedrooms	53.08 ÷ 3	**17.7 m²**	Derived: arithmetic shown
Average living room	living room floor area ÷ number of living rooms	54.47 ÷ 2	**27.2 m²**	Derived: arithmetic shown
Windows in a living room	living room windows ÷ living rooms	3 ÷ 2	**1.50**	Engine rule
Bedroom window against the room	window area ÷ room area	1.80 ÷ 17.69	**0.102**	Derived: arithmetic shown
Living room windows against the room	windows per room × window area ÷ room area	1.50 × 2.16 ÷ 27.23	**0.119**	Derived: arithmetic shown
Kitchen and bathroom against the usual minimum Advisory only; it does not change a quantity.	kitchen ≥ 5 m², bathroom with WC ≥ 2.8 m²		**Both reach the minimum**	NBC 2016 Part 3, as summarised in secondary sources; clause numbers not given
Windows against the floor area Advisory only; it does not change a quantity.	window area ≥ 10% of the floor area		**Both reach a tenth**	NBC 2016 Part 8, as summarised in secondary sources; clause numbers not given

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11. How far off the figures can be

The ranges and bands are planning judgement. They are not statistics from measured buildings.

Likely range under each total

TOTAL IN THE REPORT	FIGURE	LOW ×	HIGH ×	LIKELY RANGE
Painted area	942 m²	0.85	1.2	801 to 1,130 m²
Tiles	254 m²	0.9	1.15	228 to 292 m²
Doors and windows	22 nos	0.85	1.25	19 to 28 nos
Window glass	16 m²	0.8	1.3	13 to 21 m²

Bands for each class of figure

CLASS OF FIGURE	LOW ×	HIGH ×	BELOW (%)	ABOVE (%)	WHAT IT ALLOWS FOR
Painted area	0.85	1.2	15	20	Wall lengths come from the room plan, not from your drawings.
Putty, primer and paint	0.7	1.35	30	35	Spread varies with brand, wall roughness and how it is applied; an emulsion alone is 0.063 to 0.138 L a m² for two coats in the makers' sheets.
Tiles	0.9	1.15	10	15	Room sizes from the plan, the layout and breakage on site.
Mortar, adhesive and grout	0.65	1.3	35	30	Bed thickness and joint width vary on site.
False ceiling	0.85	1.2	15	20	Rooms and ceiling heights follow the room plan.
Doors and windows	0.85	1.25	15	25	Counts and sizes are the tool's typical ones.
Plywood and laminate	0.7	1.4	30	40	Layout of the units differs from the geometry assumed.

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Reconciliation: the report against this sheet

The totals printed in the report, rebuilt from the working above. Every row must read the same on both sides. This shows that the report and this sheet agree with each other, which is a check of the arithmetic; it does not show that either one is right against a standard or against your site.

ITEM	IN THE REPORT	FROM THIS SHEET	MATCH
Surface, net: Walls in rooms and passages	383.5 m²	383.5 m²	Yes
Surface, net: Bathroom walls above the tiles	21.6 m²	21.6 m²	Yes
Surface, net: Ceilings, inside	162.4 m²	162.4 m²	Yes
Surface, net: Outside of the building	374.4 m²	374.4 m²	Yes
Surface, net: Doors, both faces	33.2 m²	33.2 m²	Yes
Surface, net: Bathroom wall tiles	55.4 m²	55.4 m²	Yes
Surface, net: Kitchen wall above the counter, tiles	2.5 m²	2.5 m²	Yes
Surface, net: Floors to be tiled	166.9 m²	166.9 m²	Yes
Tiles to buy, Floors of rooms, with skirting	165.8 m²	165.8 m²	Yes
Tile pieces, Floors of rooms, with skirting	458 nos	458 nos	Yes
Tiles to buy, Bathroom floors	19.3 m²	19.3 m²	Yes
Tile pieces, Bathroom floors	210 nos	210 nos	Yes
Tiles to buy, Balcony floors	5.0 m²	5.0 m²	Yes
Tile pieces, Balcony floors	54 nos	54 nos	Yes
Tiles to buy, Bathroom walls	61.0 m²	61.0 m²	Yes
Tile pieces, Bathroom walls	336 nos	336 nos	Yes
Tiles to buy, Kitchen wall above the counter	2.8 m²	2.8 m²	Yes
Tile pieces, Kitchen wall above the counter	16 nos	16 nos	Yes
Tiles to order, 600 × 600 mm: pieces	458 nos	458 nos	Yes
Tiles to order, 600 × 600 mm: boxes	115 boxes	115 boxes	Yes
Tiles to order, 300 × 300 mm: pieces	264 nos	264 nos	Yes
Tiles to order, 300 × 600 mm: pieces	351 nos	351 nos	Yes
Tiles to order, 300 × 600 mm: boxes	71 boxes	71 boxes	Yes
Fixing and wet areas: Cement for the bed	2,489 kg	2,489 kg	Yes
Fixing and wet areas: Sand for the bed	4.56 m³	4.56 m³	Yes
Fixing and wet areas: Grout	31 kg	31 kg	Yes
Fixing and wet areas: Skirting	118.2 m	118.2 m	Yes
Fixing and wet areas: Waterproofing coat	26.2 m²	26.2 m²	Yes
Fixing and wet areas: Corner fillet and mesh	28.8 m	28.8 m	Yes
Stone: Door thresholds	5 nos	5 nos	Yes
Stone: Kitchen counter, granite	2.8 m²	2.8 m²	Yes
Stone: Counter edge	4.2 m	4.2 m	Yes
Paint: Wall putty, cement based	358 kg	358 kg	Yes
Paint: Primer, inside	45 L	45 L	Yes

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Paint: Interior emulsion, 2 coats	54 L	54 L	Yes
Paint: Primer, outside	37 L	37 L	Yes
Paint: Exterior acrylic emulsion	66 L	66 L	Yes
Paint: Primer, doors	3 L	3 L	Yes
Paint: Enamel or multi-surface paint, doors	3 L	3 L	Yes
Doors: number	9 nos	9 nos	Yes
Doors: shutter area	16.6 m²	16.6 m²	Yes
Doors: frame timber	0.309 m³	0.309 m³	Yes
Doors: hinges	28 nos	28 nos	Yes
Windows and glazed doors: number	13 nos	13 nos	Yes
Windows and glazed doors: glass	16.4 m²	16.4 m²	Yes
Windows and glazed doors: sections	138.5 m	138.5 m	Yes
Hardware: Door hinges	28 nos	28 nos	Yes
Hardware: Lock sets	4 nos	4 nos	Yes
Hardware: Tower bolts	5 nos	5 nos	Yes
Hardware: Handle pairs	9 nos	9 nos	Yes
Hardware: Door stoppers	9 nos	9 nos	Yes
Rooms counted in the report	9 rooms	9 rooms	Yes
Buy list: Wall putty, cement based	360 kg	360 kg	Yes
Buy list: Primer, inside	45 L	45 L	Yes
Buy list: Interior emulsion, 2 coats	54 L	54 L	Yes
Buy list: Primer, outside	38 L	38 L	Yes
Buy list: Exterior acrylic emulsion	66 L	66 L	Yes
Buy list: Primer for doors	3 L	3 L	Yes
Buy list: Enamel for doors	4 L	4 L	Yes
Buy list: Tile 600 × 600 mm	458 nos	458 nos	Yes
Buy list: Tile 300 × 300 mm	264 nos	264 nos	Yes
Buy list: Tile 300 × 600 mm	351 nos	351 nos	Yes
Buy list: Cement	50 bags of 50 kg	50 bags of 50 kg	Yes
Buy list: Sand	4.6 m³	4.6 m³	Yes
Buy list: Tile grout	32 kg	32 kg	Yes
Buy list: Waterproofing coat, bathrooms	26.2 m²	26.2 m²	Yes
Buy list: Kitchen counter, granite	2.8 m²	2.8 m²	Yes
Buy list: Counter edge, moulded	4.2 m	4.2 m	Yes
Buy list: Door thresholds	4.6 m	4.6 m	Yes
Buy list: Main door, 10 DS 21	1 nos	1 nos	Yes
Buy list: Room door, 9 DS 21	4 nos	4 nos	Yes
Buy list: Bathroom door, 8 DS 21	3 nos	3 nos	Yes
Buy list: Door of the terrace stair room, 9 DS 21	1 nos	1 nos	Yes
Buy list: Door frame timber	0.31 m³	0.31 m³	Yes
Buy list: Door hinges	28 nos	28 nos	Yes
Buy list: Lock sets	4 nos	4 nos	Yes

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Buy list: Tower bolts	5 nos	5 nos	Yes
Buy list: Handle pairs	9 pairs	9 pairs	Yes
Buy list: Door stoppers	9 nos	9 nos	Yes
Buy list: Window glass, 4 to 5.5 mm	16.4 m²	16.4 m²	Yes
Buy list: Window and sash sections	139 m	139 m	Yes
Total in the report: Painted area	942 m²	942 m²	Yes
Likely range, low end: Painted area	801 m²	801 m²	Yes
Likely range, high end: Painted area	1,130 m²	1,130 m²	Yes
Total in the report: Tiles	254 m²	254 m²	Yes
Likely range, low end: Tiles	228 m²	228 m²	Yes
Likely range, high end: Tiles	292 m²	292 m²	Yes
Total in the report: Doors and windows	22 nos	22 nos	Yes
Likely range, low end: Doors and windows	19 nos	19 nos	Yes
Likely range, high end: Doors and windows	28 nos	28 nos	Yes
Total in the report: Window glass	16 m²	16 m²	Yes
Likely range, low end: Window glass	13 m²	13 m²	Yes
Likely range, high end: Window glass	21 m²	21 m²	Yes
Band: Painted area	0.85 to 1.2	0.85 to 1.2	Yes
Band: Putty, primer and paint	0.7 to 1.35	0.7 to 1.35	Yes
Band: Tiles	0.9 to 1.15	0.9 to 1.15	Yes
Band: Mortar, adhesive and grout	0.65 to 1.3	0.65 to 1.3	Yes
Band: False ceiling	0.85 to 1.2	0.85 to 1.2	Yes
Band: Doors and windows	0.85 to 1.25	0.85 to 1.25	Yes
Band: Plywood and laminate	0.7 to 1.4	0.7 to 1.4	Yes

Standards and references used in this sheet

DOCUMENT	TITLE	USED FOR
IS 1200 (Part 11):1977	Method of measurement of building and civil engineering works: Paving, floor finishes, dado and skirting	Floors measured before the skirting and dado go on; skirting in running metres; dado in square metres. The tool follows the rules of this edition as it reads them. BIS lists a 2024 revision of Part 11, which replaces the 2013 edition, and the tool has NOT been checked against it. No clause number is quoted.
IS 1200 (Part 13):1994	Method of measurement of building and civil engineering works: White washing, colour washing, distempering and painting of building surfaces	Rules for the deduction of openings: up to 0.5 m² not taken out; from 0.5 to 3 m² one face is taken out, or half from each face when the finish differs. The tool follows the rules of this edition as it reads them. BIS lists a 2024 revision of Part 13 (it replaces the 1994 edition), and the tool has NOT been checked against it. No clause number is quoted.
IS 2395 (Part 1):1994	Code of practice for painting concrete, masonry and plaster surfaces: Operations and workmanship	Cement paint in two coats, set beside the CPWD coefficient. It gives no number of coats for emulsion or distemper, which follow the makers.
IS 1003 (Part 1):2003	Timber panelled and glazed shutters: Door shutters	Door designations (Table 3) and the hinges to a shutter (cl. 8.2). The tool applies the hinge rule to the width of the opening, not of the shutter, and takes the shutter area as the opening area (see the limits).
IS 303:2024	Plywood for general purposes: Specification	The plywood sheet of 2440 × 1220 mm. The clause number printed with it is one of the clause numbers as given in the tool's research notes; not re-read against the published text.

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IS 4021	Timber door, window and ventilator frames: Specification	Named only because it fixes frame sections. It was not read, the 100 × 65 mm section is an assumption, and no clause is quoted (clause numbers as given in the tool's research notes; not re-read against the published text).
IS 3067:1988	Code of practice for general design details and preparatory work for damp-proofing and waterproofing of buildings	Named for the waterproofing area only. Nothing is counted from it. What it covers is as given in the tool's research notes and was not re-read against the published text.
CPWD Delhi Analysis of Rates 2016	Volume 2, sub-head 13 (finishing) and Volume 1 materials	Paint, primer and exterior finish coefficients; the cement and sand in a tile bed. The item numbers printed (13.42, 13.48.2, 13.50.1) are clause numbers as given in the tool's research notes; not re-read against the published text.
Makers' data sheets	Paint, putty, adhesive, grout and tile makers	Inside paint, putty, primer and adhesive spreads; tile boxes.
NBC 2016 (Parts 3 and 8)	National Building Code of India 2016	The advisory checks on kitchen, bathroom and window sizes. The figures are as summarised in secondary sources; the Part numbers are from the tool's research notes, the code was not read, and no clause number is given.

This sheet shows the method and the arithmetic behind a planning-level quantity estimate. It lets a qualified person check the formulas and the working. It is not a structural design, a drawing or a certificate, and it does not replace the drawings and calculations that a licensed professional signs for your building.

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